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Folks Atlas

The original motivation to build a cross-chain lending protocol was to solve the fundamental issues of liquidity and availability.

- The most important factor for a lending protocol's success is liquidity. If there are stablecoins deposited, users will inevitably borrow them. By connecting to more Blockchains, the opportunity and ease at which stablecoins can be supplied increases.
- Users choose for convenience. A user wants to interact with a protocol from their favourite wallet and the chains they are most familiar with. In general a cross-chain protocol has lower barriers to entry for this reason.

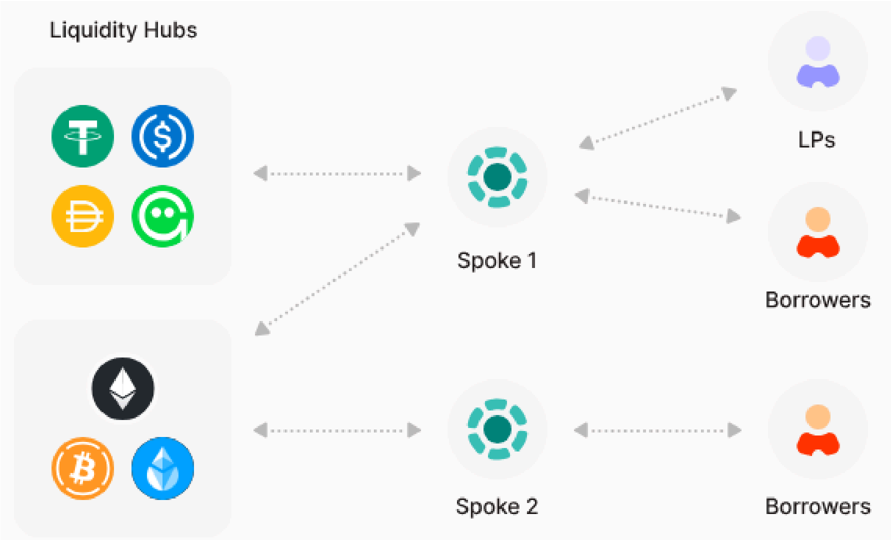
Whilst we managed to solve these issues, what we ended up with was a protocol which was often complex and slow to use. The aim of the new version of the Lending Protocol is to simplify the user experience, and thus be able to offer richer and more powerful features.

State of the Art

Here we explore the latest developments in the lending ecosystem among the major players.

AAVE v4

AAVE recently launched v4. It uses a Hub & Spoke Model (not to be confused with Hub Chain and Spoke Chains from our XChain Protocol - here the Hub & Spokes are all on the same chain).



The Liquidity Hub consolidates the tokens into single pools, while the Spokes allow borrowers to draw on the Hub Liquidity by providing a collateral.

You can think of a Spoke the same way as Folks Finance has loan types. A Spoke could be generalised or target specify tokens like stablecoins just as Folks Finance had a general loan type and a stablecoin efficiency loan type.

The main selling point AAVE has been pushing is isolated risk. Since the Spokes have caps on how much can be borrowed using them, if one Spoke had an issue then the damage it can cause is limited. Again this is the same as Folks Finance loan types which had associated caps.

The process of creating a Spoke and borrowing from a Liquidity Hub is permissioned. This means you'll need approval from the AAVE DAO.

Morpho

Morpo is the pioneer of vault based lending. It offers two separate products: Morpho Blue for variable interest rates and Morpho Midnight for fixed interest rates.

Morpho Blue

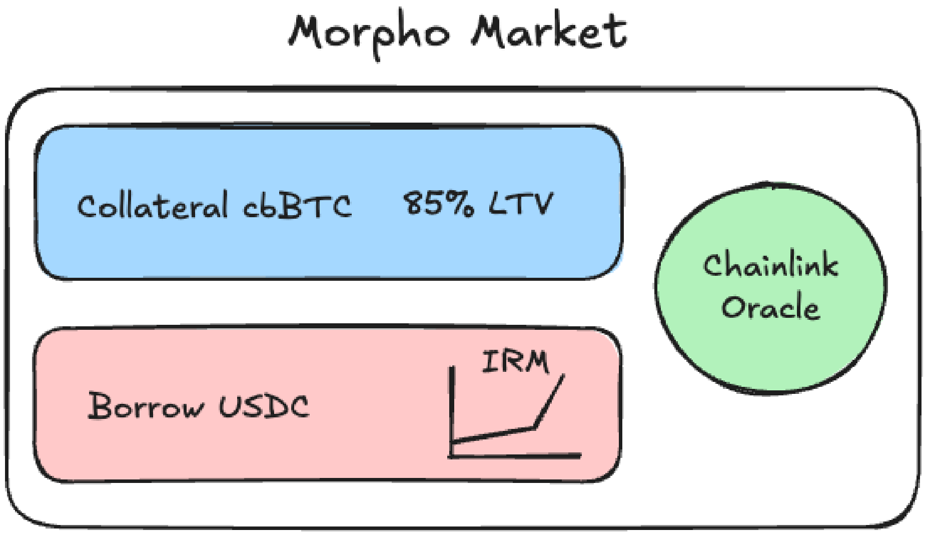
Morpho Blue offers a simple lending primitive layer that offers over-collateralised lending through the creation of permissionless immutable markets.

A **Morpho (Variable Rate) Market** is a primitive lending pool that pairs one collateral asset with one loan asset. Each market is isolated (meaning risks are contained within each individual market), immutable (cannot be changed after deployment), and will persist as long as the blockchain it deployed on is live. This design ensures predictable behavior and eliminates systematic for lenders and borrowers. Creating a Morpho Market is permissionless.

Markets follow this naming format

> *CollateralAsset/LoanAsset (LLTV%, OracleAddress, IRMAddress)* e.g,
> *wstETH/WETH (94.5%, ChainlinkOracleV2-wstETHToWETH, AdaptiveCurveIRM)*

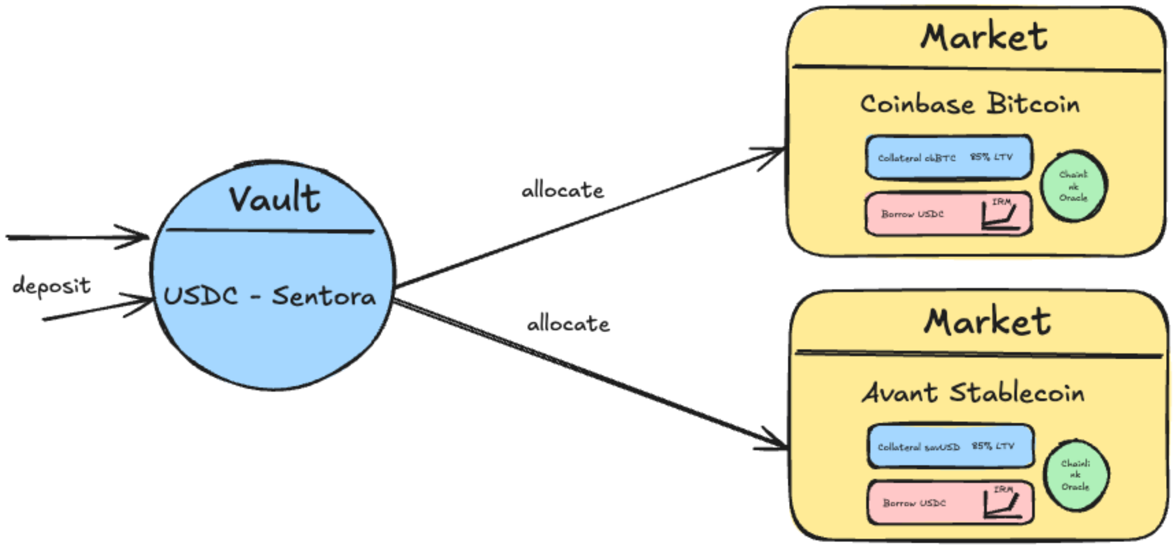
Below is a diagram showing the configuration of a Morpho Market.



Although Morpho could theoretically support any Interest Rate Model (IRM), the only IRM which has been approved for usage is the adaptive two piece-wise linear IRM.

A **Morpho Vault** is the primitive which enables depositors to lend funds to markets. You deposit a single asset (commonly a stablecoin like USDC), and the vault lends it out across a curated set of Morpho's underlying lending markets to generate yield from the interest borrowers pay. There are different roles such as an Owner, Curator, Allocator and Sentinel which each have different responsibilities for managing the Vault.

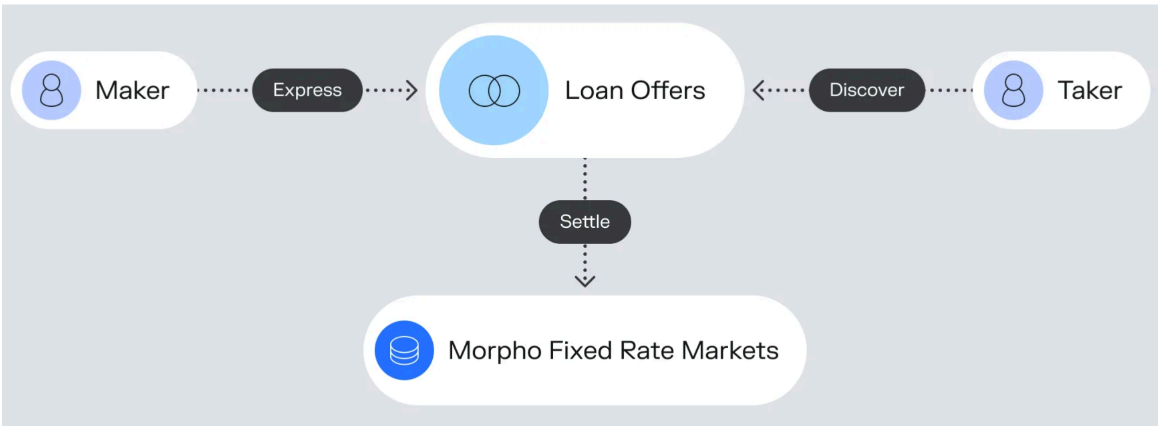
Below is a diagram showing the flow of funds from a depositor to a vault and then the various markets.



Morpho Midnight

Morpho Midnight is still in development at the time of writing.

Morpho is working on a new intent-based lending platform named Morpho Midnight. It uses fixed-rate and fixed-term loans.



The main issue with intent based lending is the UX. Either deposit makers can't earn yield immediately or borrow makers can't access liquidity immediately. To solve the deposit maker issue, Morpho Midnight will allow lenders to earn variable rates on Blue while waiting for a loan offer to be matched on Midnight.

The target demographic is likely to be more institutional focused clients who are willing to be patient for ultimately more favourable loan terms. Retail users aren't generally taking out large enough loans for there to be an active enough market for intent-based lending.

Morpho Fixed Rate Markets will support multi-collateral loans (although no support for multi-borrow loans). Morpho Vaults will be able to allocate funds to both variable rate markets with Blue and fixed rate markets with Midnight.

The New Design

In the new Folks Finance Lending Protocol, the focus is on building the infrastructure layer for lending. We will provide the tools for Curators, Asset Managers, Banks etc to create their own lending environment. These lending environments are known as Spokes (not to be confused with an XChain Spoke Chain!).

Folks Finance will leverage this tooling to offer its own lending Spoke for users but this will be second to our goal of onboarding as many stakeholders to create their own Spokes tailored to their needs.

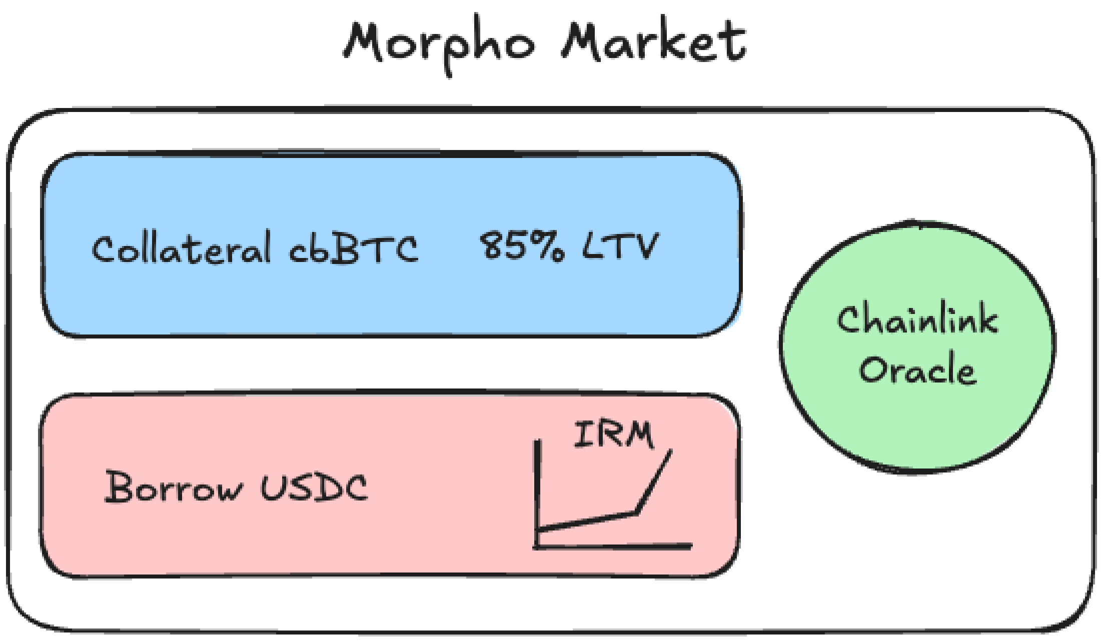
The result is a combination of the benefits of vault based lending, like Morpho, and monolithic lending, like AAVE. As a placeholder, we are naming this project **Folks Atlas**.

Spokes

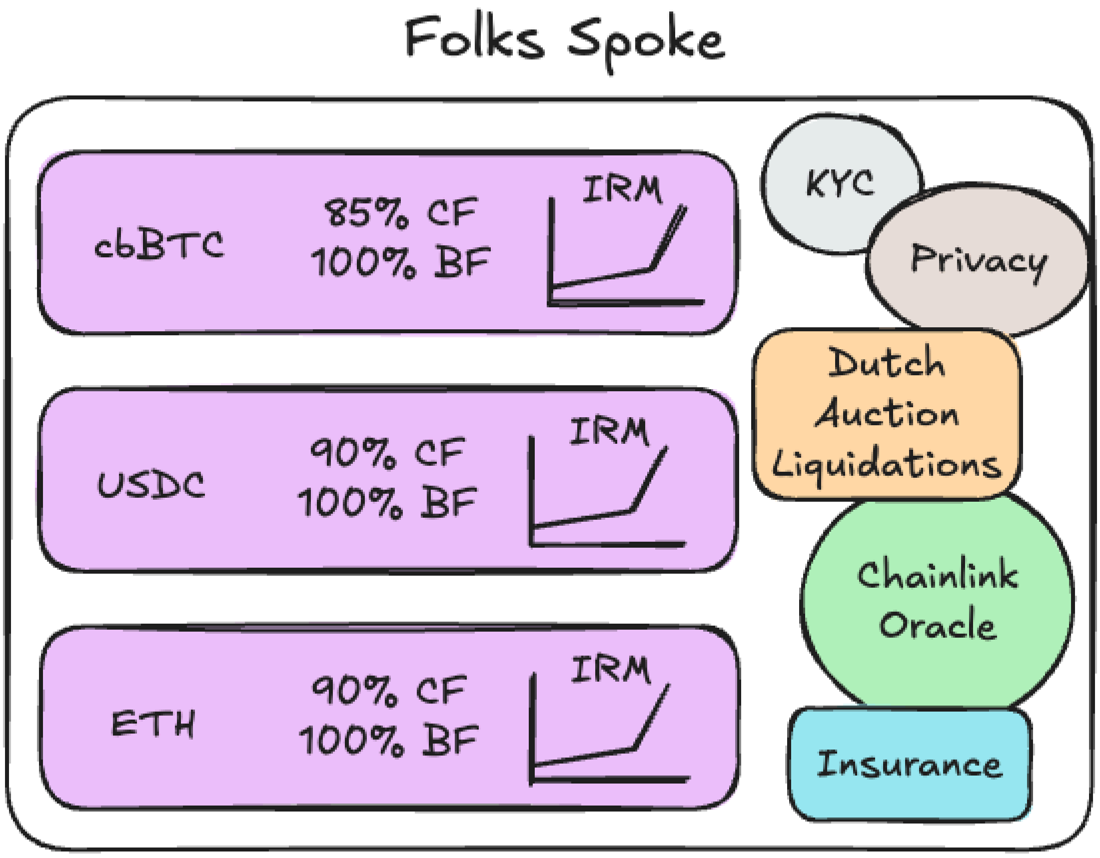
A Spoke is an isolated lending environment set up by a Curator, Asset Manager, Bank etc.

Reminder about [Morpho Markets](),

> *"A Morpho Market V1 is a primitive lending pool that pairs one collateral asset with one loan asset. Each market is isolated (meaning risks are contained within each individual market), immutable (cannot be changed after deployment), and will persist as long as the blockchain it deployed on is live. This design ensures predictable behavior and eliminates systematic risk for lenders and borrowers. Creating a Morpho Market is permissionless."*



Folks Finance Spokes supercharges the idea behind Morpho Markets. Instead of being restricted to a single collateral asset and single borrow asset, the Spoke creator is able to list as many pools as they want to support multi-collateral and multi-borrow loans.



In addition, creators can decide the interest rate model (IRM) for each pool. Then they can specify whether KYC/AML is enabled, the liquidation model to use, the oracle etc.

Below is a list of the configurable parameters:

- Chain
  - Public chains like Ethereum, Avalanche, Solana etc
  - Private chains like Kinexys, Canton etc
- Admin/Manager Roles
- Entry requirements
  - KYC/AML
  - KYT
  - Whitelisted addresses (global and per operation incl. liquidators)
- Liquidation Engine
  - Liquidation Bonus, Dutch Auction
  - Target Health Factor, Liquidation Max
  - Auto-deleverage
    - If enabled, allows use of auto-deleverage feature
- Oracle
  - Price sources such as Chainlink, Pyth etc
  - Exchange Price vs Market Price
  - Staleness, deviation, TWAP, multi-source comparison etc

- Privacy Module
    - If enabled, provides a suite of privacy solutions
- Undercollateralised Module
    - If enabled, allows setting of greater than 100% collateral factors and less than 100% borrow factors
    - Can be linked to credit scoring mechanism
- Insurance Module
    - If enabled, allows access to insurance for user losses
- Governance Module
    - If enabled, allows a token to be used for protocol listing and parameter changes
- Incentives Module
    - If enabled, allows to distribute incentives on top of Spoke
    - Integration with Merkl?
    - Referral Program
- Frontend Module
    - If enabled, allows Spoke Creator to host white label frontend customised to their brand
    - Includes support for geo-blocking
- Analytics Module
    - Risk Assessment
    - Metrics
- Security Module
    - Time lock delays
    - Monitoring
    - Alerts / Alarms
    - Rate limits
    - Automated Pausing
- Pools
    - Collaterals
        - Collateral Factor
        - Collateral Cap
        - Whether collateral is borrowable
    - Borrows
        - Borrow Factor
        - Borrow Cap
    - Interest Rate Model
        - Fixed rate, piecewise linear, double piecewise linear, exponential, piecewise exponential
        - Minimum interest accrual / pre-payment penalty
        - Origination fee - a one off fee charged at borrow time
        - Premium based on:
            - Collateral composition -> riskier assets charged higher amount
            - How close you are to being liquidated
    - Flash Loans
        - Toggle on/off
        - Can set fees and limits

Use Cases

Different use cases will require different setups and configurations. To make this easier for the Spoke Creator, we will offer Spoke Types which have default parameters so that they can onboard their project easier and in less time.

Here is an example demo flow showing the creation.

We will give an overview of each use case and how you'd configure a Spoke for them.

Generalised Lending

One use case is a generalised lending protocol e.g. AAVE v3 on Ethereum. A Spoke Creator can recreate this lending environment by adding the tokens they'd like to support and their corresponding collateral and borrow parameters.

Since a generalised Spoke is difficult to plan for, we'd categorise this under a "Custom" creation flow which includes all the available configurations and modules we have.

These Spoke's will likely be actively managed too e.g. a token's collateral factor can be reduced, a new token can be listed etc. Therefore by default these Spoke's won't be immutable.

Efficiency Lending

Another use case is when the lending protocol baskets together similarly price correlated tokens to offer higher LTV ratios. An example is the ETH efficiency loan type in Folks Finance.

One would create a Spoke for each loan type they'd want to recreate in Folks Atlas.

Again these are likely to be actively managed for the same reason as the Generalised Lending so won't be immutable by default.

Single Collateral-Borrow Pair Lending

Another use case is where the Spoke lists a single token which can be used as collateral and a single token which can be borrowed. This could be:

- Equivalent to a Morpho Blue Market where there is a token which is collateralised to borrow (most often) a stablecoin.
- An Liquid Staking Staking (LST) like wstLINK would want a single collateral wstLINK and a single borrow LINK for easy looping.
- A yield bearing stablecoin.

Permissioned Lending

An institution like a Bank generally avoids open protocols where the participants in a market are unknown. They'd prefer to work in a closed off environment where they have control and visibility as to who's involved.

Therefore for these use cases, we'd allow them to create a Spoke with access control restrictions such as KYC or whitelists. They'd also be privacy features including support to deploy in a private blockchain.

Private Lending

Similar to Permissioned Lending with a subtle difference, Private Lending would be two parties who have made an off-chain agreement on loan terms and they'd like to formalise and execute the loan on-chain.

For these Spokes, there would be access control where a single address is allowed to borrow from the Spoke. Also there would be the option for under-collaterised lending here because of the existence of the off-chain agreement.

Intent Based Lending

The last use case we will discuss is intent based lending where the lender and borrower agree on fixed terms and rates for a position. Think the equivalent to Morpho Midnight Market.

These can be set up by an intent-based lender (maker) setting up their own Spoke with the parameter's they'd be willing to lend under. For an intent-based borrower (maker), we'd need a way to store and permissionlessly execute this intent once a suitable Spoke is created.

RWA Lending

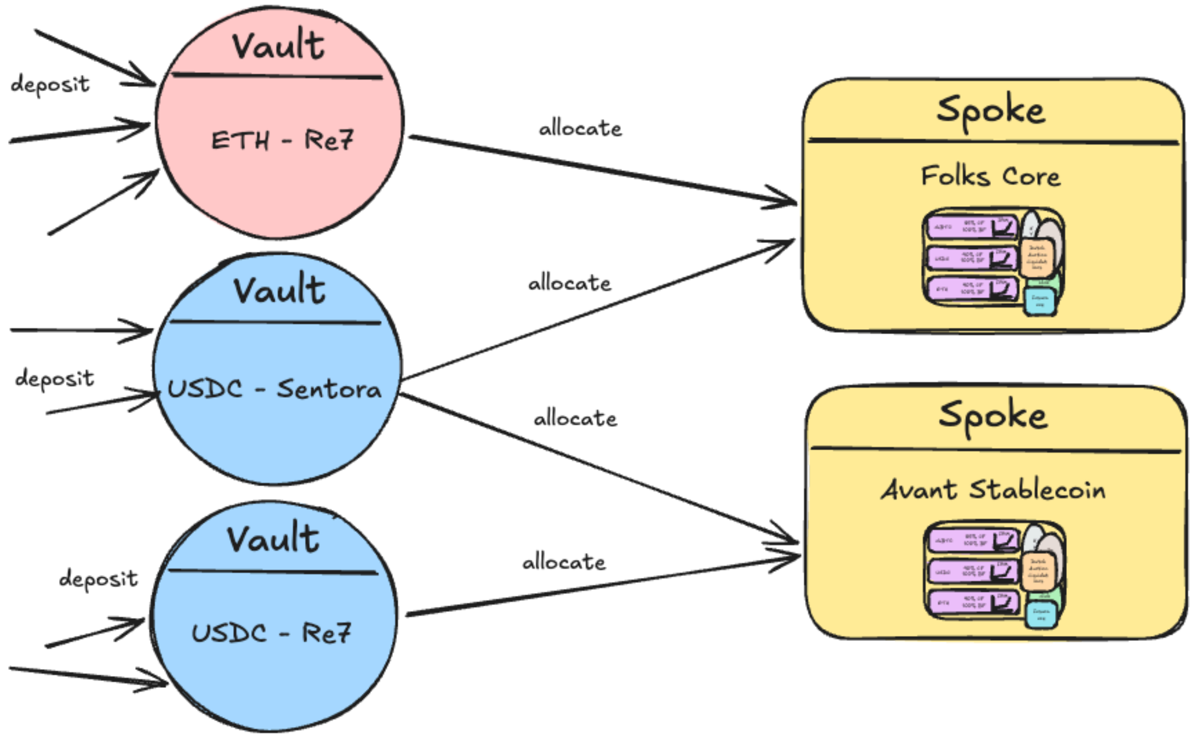
We can enable issuers like Ondo, Superstate, Maple, etc to deploy their own lending environments around their Real World Assets (RWAs).

We would need to investigate what exactly their needs are and then tailor a Spoke configuration which would be the right fit for them.

Vaults

A Vault is a smart contract which users can deposit into, where a Curator allocates the liquidity available in the vault across the different Spokes.

If you are familiar with Morpho Vaults, the concept is similar. Where Spokes provides the raw infrastructure for lending, most end users do not want to evaluate individual collateral/borrow pairings, interest rate models, oracles, or liquidation parameters. Vaults abstract this complexity away, giving depositors a single, passive entry point while delegating risk assessment and yield optimisation to a Curator.



A Curator packages a strategy — a set of approved Spokes, allocation targets, caps, and rebalancing logic — which depositors can access with a single deposit. The result is that a user who simply wants to earn yield on USDC can do so without ever needing to understand the underlying Spoke configuration, while a more sophisticated user retains the option to interact with Spokes directly.

Below is a list of the configurable parameters (a lot of these are shared with Spokes so see Spokes for further explanation):

- Chain
- Token
- Admin/Manager Roles
- Entry requirements
- Privacy Module
- Insurance Module
- Incentives Module
- Frontend Module
- Analytics Module
- Security Module

Cross-Chain Support

In our XChain protocol, it was possible to deposit on one chain and borrow on another chain. However this feature wasn't used often; users' strategies mainly focused on tokens available on the same chain. Therefore we will be abandoning this as well as all the complex account management and cross-chain messages which were needed to support it.

It's undecided at this stage to what extent the new protocol will support cross-chain liquidity. We could either:

a. Integrate it directly into the protocol itself e.g. cross-chain USDC pools where liquidity is bridged using Circle CCTP and this is hidden from the user.
b. Integrate it in the frontend where the user can bridge USDC after receiving it in their wallet.
c. Externalise support where it's up to the user to later bridge the tokens if they'd like.

Realistically the most practical solution would be to support cross-chain vaults whilst keeping Spokes strictly single chain. That way the Vault Allocator could move liquidity to where there is a borrow demand.

Frontend

There are a number of different frontends we will have depending on who the stakeholder is.

We must have:

- An admin dashboard to allow the creation and management of a Spoke. Example demo flow showing the creation. Note that this is specifically unique to Folks Atlas. With Morpho, a market is something which is very basic and immutable so there is nothing to manage.
- An admin dashboard to allow the creation and management of a Vault.
- A retail focused website which allows users to deposit into Vaults and access Loans across a range of Spokes. Example demo retail dapp.
- A retail focused white-label style website where projects can link solely to their own Vaults and Spokes. This would allow a project to advertise their opportunities without having to compete with other providers on a website which advertises everyone.MVP

The following tables give a high-level overview of what is needed. We need to dive deeper into some of these to scope feasibility, different approaches and the work required. Note "DD" stands for "Design Decision".

Spoke Dashboard

Title	MVP	Description
Connect Wallet	Yes	A curator can connect their wallet in order to issue transactions.
Multisig Connection	No	Use Safe Wallet for multisig deployment and management
Basic Create Spoke	Yes	A curator should have the ability to create a basic spoke. • Chain

		• Admin/Manager roles
		• Liquidation Engine
		• Oracle
		  ○ Price sources for each supported token (Chainlink, Pyth, custom)
		  ○ Adapter contracts
		  ○ YT-PT pendle oracle
		  ○ Market vs Exchange Rate
		• Pools
		  ○ IRM
		  ○ Caps
		  ○ Collateral and borrow factors
		  ○ Flash loans supported
		  ○ Whether collateral is borrowable
		• Fee parameters
		  ○ Percentage of interest retained for each supported token

Whitelisted Spoke Entry Requirements	Yes	A curator can configure their spoke such that only specified addresses can do certain actions.
KYC/AML Spoke Entry Requirements	No	A curator can configure their spoke to require KYC/AML checks before a user can interact with the spoke.
Private Spoke	No	A curator can configure their spoke to be private. • Can deploy to a private blockchain • Can add privacy on top of public blockchain deployment
Undercollateralised Spoke	No	A curator can enable undercollateralised lending in their spoke. • Configure parameters • Likely would require “Whitelisted Spoke Entry Requirements” to be enabled • Alternative would be link to credit scoring mechanism
Immutable Spoke	Yes	A curator can create an immutable Spoke, either: • At creation. • After creation by renouncing all roles.
Insurance for Spoke	No	A curator can add insurance to their spoke to protect against bad debt and/or hacks.
Whitelabel Frontend	No	A curator can deploy their own custom frontend under their domain which enables users to interact with their spoke with no competitors visible.
Governance Module	No	A curator can execute Spoke parameter decisions through governance.
Incentives Module	No	A curator can distribute incentives on top of their Spoke.
Security Module Parameters	Yes	A curator can configure security parameters. • Time lock delays on important admin/manager operations. • Rate limits on withdraw outflows, collateral inflows, borrow outflows
Security Module Monitoring	No	A curator can configure on-chain monitoring / alarms.
Manage Spoke	Yes	A curator should have the ability to manage their existing spokes including updating parameters, modules etc. • There are some things which cannot change once you deploy a Spoke e.g. the chain.
Spoke Analytics	No	A curator should be able to view analytics relating to their spoke such as total users, TVL, interest rates, liquidations, oracle feed status, etc.

Vault Dashboard

Title	MVP	Description
Connect Wallet	Yes	A curator can connect their wallet in order to issue transactions.
Multisig Connection	No	Use Safe Wallet for multisig deployment and management
Basic Create Vault	Yes	A curator should have the ability to create a basic vault. ● Chain ● Token ● Admin/Manager roles
Queue Management	Yes	A curator needs to be able to adjust supply and withdraw queues. ● Supply cap per spoke ● Manual reallocation deposit per spoke ● Automatic rebalancing
Removing Process	Yes	A curator can remove a Spoke from a Vault. ● Mark Spoke for removal (auto-prioritized in withdrawal queue, removed from supply queue)
KYC/AML Spoke Entry Requirements	No	A curator can configure their vault to require KYC/AML checks before a user can interact with the vault.
Privacy Vault	No	
Insurance for Vault	No	
Incentives Module	No	
Security Module Monitoring	No	A curator can configure on-chain monitoring / alarms.
Vault Analytics	No	A curator should be able to view analytics relating to their Vault such as total users, TVL, interest rates, etc.

Retail DApp

Title	MVP	Description
Connect Wallet	Yes	A user can connect their wallet in order to issue transactions.
User Dashboard	Yes	A user can see a dashboard overview of all their positions across Spokes. ● Gamified information such as live interest accrual. ● Historical transactions.
View Spokes	Yes	A user can view, search and filter for Spokes. There is also basic information such as the deployed chain and TVL.
Highlighted Spokes	Yes	Folks Finance can prioritise certain Spokes to appear in more prominent placements.
Select Spoke	Yes	A user can click into a Spoke's dedicated page. This page should also be linkable directly.
Deposit/Withdraw in Spoke	DD	A user can deposit and withdraw directly into Spoke. The other possibility is that the only way to deposit into Spoke is through a Vault.
Multiple Positions	DD	A user can manage multiple positions within a Spoke.

Position Health	Yes	A user can see the health of their Spoke position e.g. how close they are to being liquidated.
Position Alerts	No	A user can subscribe to be alerted when a position meets certain conditions such as below X health factor or above Y interest rate.
Auto Deleverage	No	A user can have their positions automatically de-leveraged with no liquidation penalty when a position meets certain conditions such as below X health factor.
Collateralise in Spoke	Yes	A user can add and remove collateral in their Spoke position.
Borrow in Spoke	Yes	A user can borrow and repay in their Spoke position.
Spoke Security Assessment	No	A user can read a risk/security assessment of a Spoke in order to make informed decisions

Features

Some above titles link to a bookmarked section when they require further information.

KYC/AML Spoke Entry Requirements

We need to find and confirm a partner to launch this feature with so we can determine what's feasible. Ideally we have the following:

Title	Location	Description
Signup	Curator Dashboard + KYC Provider	A curator can set up an account with the KYC Provider. This could be done through a referral link.
Link	Curator Dashboard	A curator needs to be able to connect their KYC Provider account to their Curator Dashboard.
User requirements	KYC Provider	A curator can specify their KYC requirements e.g. ● Valid passport ● Not from restricted countries ● Has bank account ● etc
Manage users	KYC Provider	A curator can review applications and approve/reject individual users on a case-by-case basis if needed.
User KYC	Retail DApp + / KYC Provider	A user can complete the KYC process for a Spoke.

Private Spoke

TBD

Insurance for Spoke

TBD

Whitelabel Frontend

Ideally we have the following:

Title	Location	Description
Hosting	Curator Dashboard	A curator can launch their own custom Spoke frontend at their given domain.
Spoke Branding	Curator Dashboard + Whitelabel Frontend	A curator can use their own logo and colour scheme.
Updates	Curator Dashboard + Whitelabel Frontend	A curator can update their frontend to get bug fixes / new features which the Folks Finance team has made.
Public Github Code	Github	A curator can make a PR to fix or add new features to the Folks Finance Github repo.
Custom Code	Github	A curator can fork the Folks Finance Github repo and edit the code to add their own custom features / needs.
Retail	Whitelabel Frontend	A user can interact with the Spoke in the same manner as they could in the shared global frontend.

Governance Module

TBD

Incentives Module

We need to speak with Merkl. Ideally we have the following:

Title	Location	Description
Launch	Curator Dashboard + Merkl	A curator can launch a new incentive campaign for their spoke.
Lending	Merkl	A curator can distribute incentives to net lending of specific tokens.
Borrowing	Merkl	A curator can distribute incentives to net borrowing of specific tokens.
Custom Actions	Merkl	A curator can distribute incentives for certain actions such as looping.

Security Module Monitoring

We need to speak with on-chain monitoring solutions like Hypernative. Ideally we have the following:

Title	Location	Description
Signup	Curator Dashboard + Hypernative (or similar)	A curator can set up an account with the on-chain monitoring solution. This could be done through a referral link.
Alerts	Hypernative (or similar)	A curator can configure custom alerts for anomalous behaviour. Alerts can be received via email, slack, discord etc.
Automated Pausing	Curator Dashboard + Hypernative (or	A curator can automatically pause their contracts based on an alert.

	similar)	

Stablecoin

TODO

Revenue

Split retention rate fee
Split vault performance fee
One time setup fee
Staking Folks to keep Spoke / Vault open
Partner referrals

EIPs

Tokenized Vaults https://eips.ethereum.org/EIPS/eip-4626
Permit Extension https://eips.ethereum.org/EIPS/eip-2612
Flash Loans https://eips.ethereum.org/EIPS/eip-3156
Standard Interface Detection https://eips.ethereum.org/EIPS/eip-165
Common Quote Oracle https://eips.ethereum.org/EIPS/eip-7726
Transient storage https://eips.ethereum.org/EIPS/eip-1153

Patterns

Execution Context (decoupling the auth user from the msg.sender)
Batch transactions
Proxies to reduce deployment costs. We would deploy the implementation of pool / spoke / modules first and then they are just deploying a proxy set to one of those implementations (so they can have their own state). The proxy pattern here is to reduce gas costs, the proxies aren't upgradeable

Appendix / Notes

- Global pool with shared liquidity, every token cross-chain with NTT or token specific bridge (like CCTP for USDC)
- They can ask for approval to tap into global liquidity. We set caps for Spoke. Potentially we can also approve fixed interest rate borrowing for Spoke.
- Spoke will list their own tokens too. The Curator can decide if the collaterals can be borrowed and if they go to the shared liquidity pool.
- When an everyday user deposits, they generally do this in shared liquidity. However Curator could share a link where the user is depositing into their Spoke (or through their Spoke into the shared liquidity pool).

How Morpho does KYC
https://docs.morpho.org/curate/concepts/gates. Another option they describe is for the token itself to implement access control restrictions.

Silo allows user to decide if they want their collateral to be borrowable
https://docs.silo.finance/docs/developers/protocol-overview/architecture#silos-share-tokens

Hooks
https://github.com/silo-finance/silo-contracts-v3/blob/develop/silo-core/docs/Hooks.md

AI Docs
https://docs.morpho.org/tools/ai/llms/